

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO2	Assessment:	Assessment Tool 2 – Scenario-Based Requirement Analysis
Weightage:	20%	Total Marks:	25
CO2 Statement:	<i>Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills</i>		
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Date:	03/08/2026	Duration:	60 minutes

Code of Conduct

*I **Y. Ramadevi** - 192373065 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: **Y. Ramadevi**

Online Food Delivery Management System

1. Study and Analyse the Scenario

1.1 Problem Statement

The **Online Food Delivery Management System** is a software application that enables customers to order food online from nearby restaurants through a web or mobile application. The system allows customers to browse restaurant menus, place orders, make secure online payments, and track delivery status in real time. Restaurants receive and process orders, while delivery partners deliver the food to customers. The administrator manages restaurants, users, orders, payments, offers, complaints, and system reports.

The current manual ordering process often leads to delayed deliveries, incorrect orders, poor communication between customers and restaurants, payment issues, and inefficient order tracking. The proposed system aims to automate the complete food ordering and delivery process while improving customer satisfaction and operational efficiency.

1.2 Business Domain

The proposed system belongs to the **Food Delivery and E-Commerce** domain. It integrates restaurants, customers, delivery partners, payment gateways, and administrators into a single online platform for efficient food ordering and delivery.

1.3 Stakeholders

The major stakeholders involved in the system are:

- Customers
- Restaurant Owners/Managers
- Delivery Partners
- System Administrator
- Payment Gateway Provider
- Restaurant Staff
- Customer Support Team

1.4 Existing Challenges

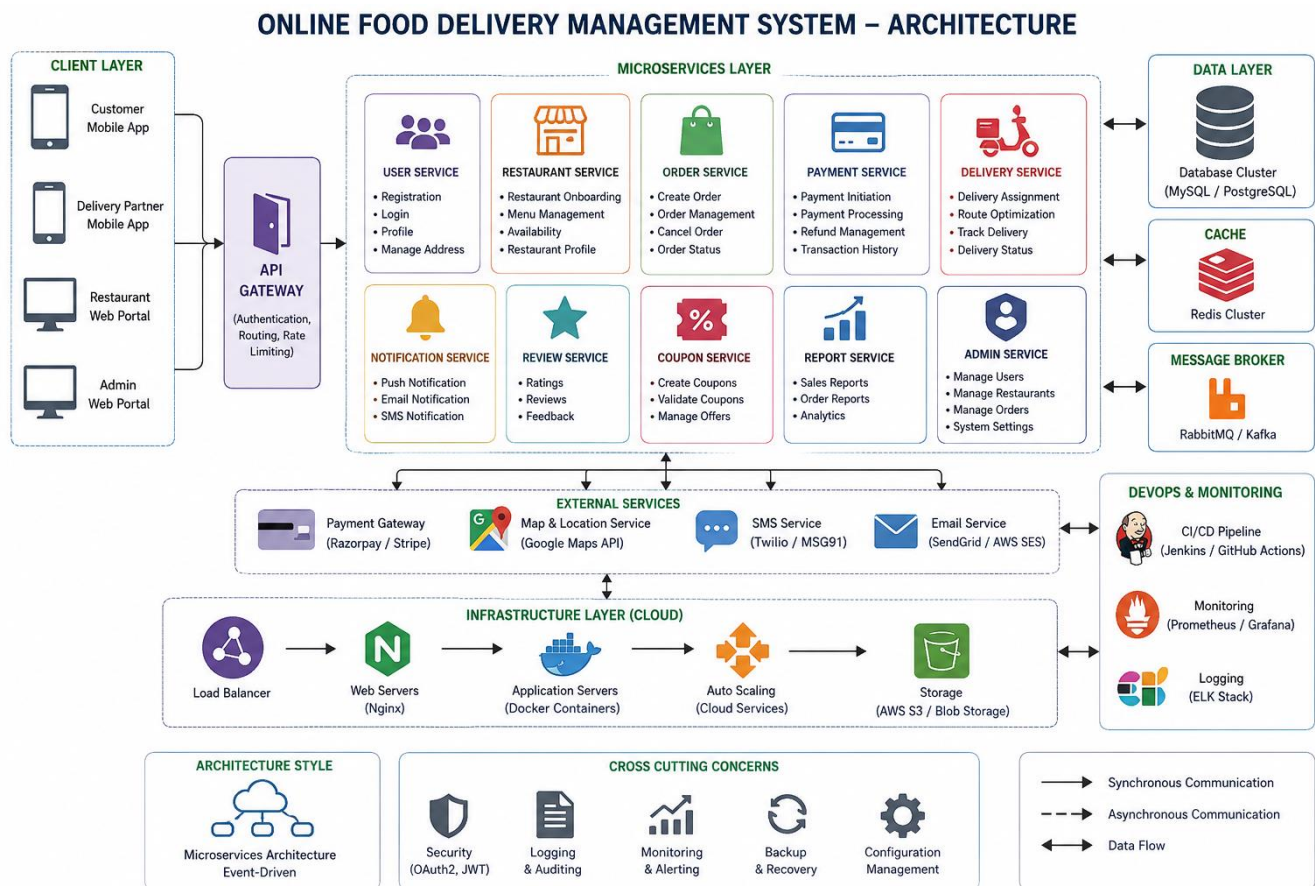
The existing food ordering system faces several challenges:

- Manual order processing causes delays.
- Incorrect order entries lead to customer dissatisfaction.
- Difficulty in tracking delivery status.
- Limited communication between customers and restaurants.
- Lack of secure online payment options.
- Inefficient management of restaurant menus.
- High chances of duplicate or missing orders.
- No centralized database for order history.
- Poor customer feedback management.
- Difficulty in managing multiple restaurants simultaneously.

1.5 Objectives of the Proposed System

The objectives of the Online Food Delivery Management System are:

- Automate the food ordering process.
- Provide real-time order tracking.
- Ensure secure online payment processing.
- Improve communication between customers and restaurants.
- Reduce order processing time.
- Increase customer satisfaction.
- Maintain accurate order records.
- Improve restaurant management efficiency.
- Provide reliable delivery management.
- Generate business reports for decision-making.



2. Functional and Non-Functional Requirements

2.1 Functional Requirements

The system should provide the following functionalities:

Customer Module

- User Registration
- User Login
- Forgot Password
- Browse Restaurants

- Search Food Items
- Filter Restaurants
- View Food Menu
- Add Items to Cart
- Update Cart
- Remove Items from Cart
- Place Order
- Online Payment
- Cash on Delivery Option
- Track Order
- Cancel Order
- View Order History
- Submit Ratings and Reviews
- Contact Customer Support

Restaurant Module

- Restaurant Registration
- Restaurant Login
- Manage Food Menu
- Update Food Availability
- Accept or Reject Orders
- Update Order Status
- View Sales Reports

Delivery Partner Module

- Delivery Partner Login
- Accept Delivery Requests
- View Assigned Orders
- Update Delivery Status
- Confirm Delivery
- View Earnings

Administrator Module

- Manage Customers
- Manage Restaurants
- Manage Delivery Partners
- Manage Orders
- Manage Payments
- Manage Offers and Coupons
- Manage Complaints
- Generate Reports
- Monitor Overall System Performance

2.2 Non-Functional Requirements

Performance

- System should process customer requests within **2–3 seconds**.
- Support thousands of concurrent users.

Security

- Secure login authentication.
- Password encryption.
- SSL-enabled payment gateway.
- Role-based access control.

Reliability

- System availability should be **99.9%**.
- Automatic backup and recovery.

Usability

- User-friendly interface.
- Easy navigation.
- Mobile-responsive design.

Scalability

- Support increasing users and restaurants.
- Easy integration with new payment gateways.

Maintainability

- Modular software architecture.
- Easy software updates and bug fixes.

Availability

- System available **24×7**.
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3. Identify Actors and Use Cases

Primary Actors

- Customer
- Restaurant Manager
- Delivery Partner
- Administrator

Secondary Actors

- Payment Gateway
 - Notification Service
 - GPS Service
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Major Use Cases**Customer**

- Register
- Login
- Browse Restaurants
- Search Food
- View Menu
- Add to Cart
- Place Order
- Make Payment
- Track Order
- Cancel Order
- View Order History
- Rate Food

Restaurant Manager

- Login
- Manage Menu
- Accept Orders
- Reject Orders
- Update Order Status

- View Sales Report

Delivery Partner

- Login
- Accept Delivery
- Navigate to Customer
- Deliver Food
- Update Delivery Status

Administrator

- Manage Users
- Manage Restaurants
- Manage Delivery Partners
- Monitor Orders
- Manage Payments
- Generate Reports
- Manage Offers

Use Case Diagram (Text Representation)

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+-----+
| Online Food Delivery System |
+-----+

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Customer -----> Register
Customer -----> Login
Customer -----> Browse Restaurants
Customer -----> Search Food
Customer -----> Add to Cart
Customer -----> Place Order
Customer -----> Make Payment
Customer -----> Track Order
Customer -----> Give Review

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Restaurant Manager -----> Manage Menu
Restaurant Manager -----> Accept Orders
Restaurant Manager -----> Update Order Status

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Delivery Partner -----> Accept Delivery
Delivery Partner -----> Deliver Food
Delivery Partner -----> Update Delivery Status

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Administrator -----> Manage Users
Administrator -----> Manage Restaurants
Administrator -----> Manage Orders
Administrator -----> Generate Reports

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Payment Gateway -----> Process Payment

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Notification Service -----> Send Notifications

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4. Software Requirement Specification (SRS)

4.1 Introduction

The Online Food Delivery Management System provides an integrated platform for customers, restaurants, delivery partners, and administrators. The system automates ordering, payment, and delivery processes to improve efficiency and customer satisfaction.

4.2 Problem Description

Traditional food ordering methods are slow, error-prone, and difficult to monitor. Customers often face delays, incorrect orders, and limited tracking facilities. Restaurants also experience challenges in order management and customer communication.

4.3 Scope of the System

The system supports:

- Customer registration
 - Restaurant management
 - Online ordering
 - Online payment
 - Delivery management
 - Order tracking
 - Feedback management
 - Report generation
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4.4 Functional Requirements

- User Management
 - Restaurant Management
 - Food Ordering
 - Payment Processing
 - Delivery Tracking
 - Feedback System
 - Reporting Module
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4.5 Non-Functional Requirements

- High Performance
 - High Security
 - Scalability
 - Reliability
 - Availability
 - Maintainability
 - Easy Usability
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4.6 Use Case Descriptions

Use Case	Description
Register	Customer creates a new account
Login	User accesses the system
Browse Menu	Customer views available food items
Place Order	Customer confirms food order
Make Payment	Payment through online gateway
Track Order	Customer monitors delivery status

Manage Menu	Restaurant updates food menu
Deliver Order	Delivery partner delivers food

4.7 Data Requirements

The database should store:

- Customer Details
 - Restaurant Details
 - Menu Details
 - Orders
 - Payments
 - Delivery Information
 - Reviews
 - Coupons
 - Reports
-

4.8 External Interface Requirements

User Interface

- Responsive Web Interface
- Android/iOS Mobile Application

Hardware Interface

- GPS Device
- Mobile Phones
- POS Systems

Software Interface

- Payment Gateway API
 - Google Maps API
 - SMS Gateway
 - Email Service
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4.9 System Features

- Online Registration
 - Restaurant Search
 - Smart Food Search
 - Cart Management
 - Secure Payment
 - Live Order Tracking
 - Push Notifications
 - Restaurant Dashboard
 - Delivery Dashboard
 - Admin Dashboard
 - Ratings & Reviews
 - Offers & Coupons
 - Analytics Reports
-

5. Assumptions and Constraints

5.1 Assumptions

- Users have internet connectivity.

- Customers provide valid delivery addresses.
 - Restaurants update menus regularly.
 - Delivery partners use GPS-enabled smartphones.
 - Payment gateways are available.
 - Users possess basic digital literacy.
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5.2 Constraints

Technical Constraints

- Stable internet connection required.
- Third-party API dependencies.

Operational Constraints

- Restaurant operating hours.
- Delivery partner availability.

Economic Constraints

- Development and maintenance costs.
- Payment gateway transaction fees.

Legal Constraints

- Data privacy regulations.
- Secure online payment compliance.

Time Constraints

- Timely delivery of food.
 - System implementation schedule.
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6. Requirement Validation

6.1 Completeness

All stakeholder requirements including customer, restaurant, delivery partner, administrator, and payment gateway have been identified.

6.2 Consistency

Requirements are consistent and do not conflict with the system objectives.

6.3 Ambiguity Check

Each requirement is written clearly to avoid multiple interpretations.

6.4 Redundancy Check

Duplicate requirements have been removed to improve clarity.

6.5 Missing Requirements

Future enhancements such as AI-based food recommendations, voice ordering, drone delivery, and multilingual support can be added.

6.6 Suggested Improvements

- Integrate AI-based food recommendations.
- Provide chatbot-based customer support.
- Enable real-time traffic-aware delivery routing.
- Support multilingual user interfaces.
- Introduce loyalty and reward programs.

- Implement advanced analytics dashboards for restaurants and administrators.
- Add contactless delivery and digital wallet support.

Conclusion

The **Online Food Delivery Management System** streamlines the entire food ordering and delivery process by connecting customers, restaurants, delivery partners, and administrators through a unified platform. By clearly identifying functional and non-functional requirements, defining stakeholder interactions, and preparing a comprehensive Software Requirement Specification (SRS), the system ensures reliability, scalability, security, and user satisfaction. This analysis provides a strong foundation for developing a robust, efficient, and maintainable food delivery application.